

Name

M#

Exam Tuesday

Models

Linear Algebra Review

We are going to learn how to write down the general solution of

$$y' = A y$$

from the eigenvalues and eigenvectors of the matrix A .

Just a reminder

- $A v_i = \lambda_i v_i$ with $\|v_i\| \neq 0$ defines an eigenvalue λ_i and associated eigenvector v_i .
- Eigenvalues and eigenvectors can be complex.
- Almost all $n \times n$ matrices have an eigenvector basis $\{v_1, v_2, \dots, v_n\}$. $V = [v_1 \mid v_2 \mid \dots \mid v_n]$ gives $V^{-1} \cdot A \cdot V = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$. This is diagonalization.

```
In[ ]:= n = 4;  
A = RandomReal[{-1, 1}, {n, n}];  
{λs, V} = Eigensystem[A]; V = V^T;  
λs  
MatrixForm[Inverse[V].A.V]  
  
Out[ ]:=  
{-0.900268, 0.774685, -0.12763, 0.0371144}
```

```
Out[ ]//MatrixForm=  

$$\begin{pmatrix} -0.900268 & -8.88178 \times 10^{-16} & 5.55112 \times 10^{-16} & -5.55112 \times 10^{-16} \\ -2.22045 \times 10^{-16} & 0.774685 & -1.38778 \times 10^{-16} & 1.66533 \times 10^{-16} \\ -6.73073 \times 10^{-16} & 2.7929 \times 10^{-16} & -0.12763 & 3.81639 \times 10^{-17} \\ -7.42462 \times 10^{-16} & 2.08167 \times 10^{-16} & 8.32667 \times 10^{-17} & 0.0371144 \end{pmatrix}$$

```

Linear Algebra Review and ODE $y' = A y$

Suppose I have $A v_1 = \lambda_1 v_1$ then $y_1(t) = c_1 e^{\lambda_1 t} v_1$ solves $y' = A y$.

- All the eigen-pairs $A v_i = \lambda_i v_i$ give solutions $y_i(t) = c_i e^{\lambda_i t} v_i$.
- $y' = A y$ is linear so $y(t) = \sum_{i=1}^n c_i e^{\lambda_i t} v_i$ is an n parameter family of solutions.

- If the eigen-vectors are Linearly Independent (LI) then there are unique coefficients $c = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$

satisfying $y(0) = \sum_{i=1}^n c_i e^{\lambda_i 0} v_i = y_0$. All we do is solve $Vc = y_0$.

- Summary. The solution of $y' = Ay$ with $y(0) = y_0$ is $y(t) = \sum_{i=1}^n c_i e^{\lambda_i t} v_i$ with $Vc = [v_1 \mid \dots \mid v_n]c = y_0$
- You can rewrite a second order ODE as a first order system.

$Li' + Ri + q/C + 12 = 0$ with $i = q'$ is the same as

$$\begin{pmatrix} q' \\ i \end{pmatrix} = \begin{pmatrix} q' \\ -\frac{q}{CL} - \frac{i}{RL} - 12 \end{pmatrix} = \begin{pmatrix} i \\ -\frac{q}{CL} - \frac{i}{RL} - 12 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{1}{CL} & -\frac{1}{RL} \end{pmatrix} \begin{pmatrix} q \\ i \end{pmatrix} + \begin{pmatrix} 0 \\ -12 \end{pmatrix}$$

- We need to know how to think about complex eigenvalues $\lambda = a \pm bi$.

- We need to know what $e^{a+bi} = e^a e^{bi}$ means.

- Starts of Series

$$e^t = 1 + t + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{24} + \frac{t^5}{120} + \frac{t^6}{720} + \frac{t^7}{5040} + \frac{t^8}{40320} + O[t]^9$$

$$\sin(t) = t - \frac{t^3}{6} + \frac{t^5}{120} - \frac{t^7}{5040} + O[t]^9$$

$$\cos(t) = 1 - \frac{t^2}{2} + \frac{t^4}{24} - \frac{t^6}{720} + \frac{t^8}{40320} + O[t]^9$$

- Full Series

$$e^t = \sum_{n=0}^{\infty} \frac{1}{n!} t^n$$

$$\sin(t) = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)!} t^{2j+1}$$

$$\cos(t) = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j)!} t^{2j}$$

$$E^{bi} = \sum_{n=0}^{\infty} \frac{1}{n!} (bi)^n = 1 + ib - \frac{b^2}{2} - \frac{ib^3}{6} + \frac{b^4}{24} + \frac{ib^5}{120} - \frac{b^6}{720} - \frac{ib^7}{5040} + \frac{b^8}{40320} + O(b^9)$$

$$E^{bi} = \left(1 - \frac{b^2}{2} + \frac{b^4}{24} - \frac{b^6}{720} + \frac{b^8}{40320} + \dots\right) + i \left(b - \frac{b^3}{6} + \frac{b^5}{120} - \frac{b^7}{5040} + \dots\right)$$

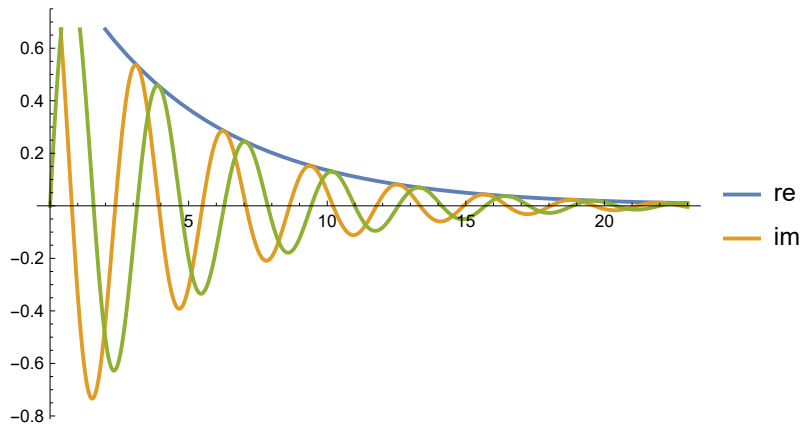
$$E^{bi} = \cos(b) + i \sin(b)$$

$$e^{(a \pm bi)t} = e^{at} (\cos(bt) \pm i \sin(bt)).$$

- Solutions grow or decay according to e^{at}
- Solutions wiggle like $\sin(bt)$ or $\cos(bt)$

```
In[19]:= {a, b} = {-0.2, 2};
Plot[{E^a t, Re[e^(a+b i) t], Im[e^(a+b i) t]}, {t, 0, 23},
PlotLegends -> {"re", "im"}]
```

Out[20]=



```
In[26]:= A =  $\begin{pmatrix} -1. & 6 & 1 \\ -6 & -2 & 1 \\ 1 & 1 & -5 \end{pmatrix}$ ;
```

Eigenvalues[A]

```
y0 = {7, 4, 5}; TMax = 3;
```

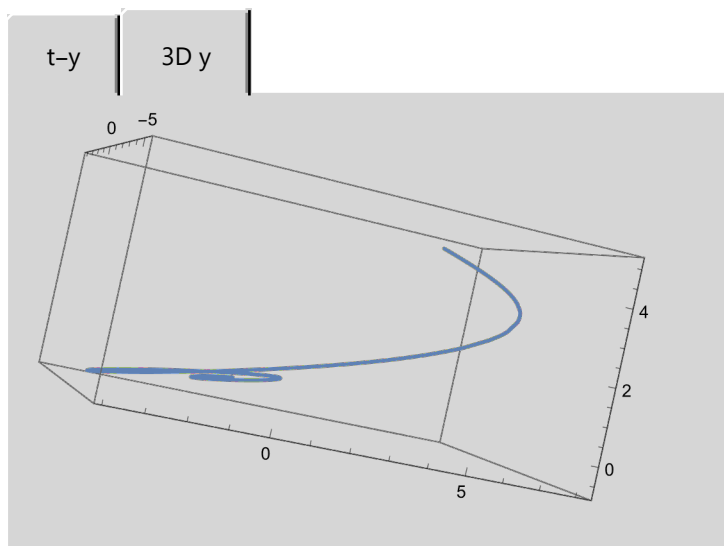
```
ySol = NDSolveValue[{y'[t] == A.y[t], y[0] == y0}, y, {t, 0, TMax}];
```

```
TabView[{
  "t-y" -> Plot[ySol[t], {t, 0, TMax}, PlotRange -> All],
  "3D y" -> ParametricPlot3D[ySol[t], {t, 0, TMax}, PlotRange -> All]
}]
```

Out[27]=

```
{-1.42563 + 5.85552 i, -1.42563 - 5.85552 i, -5.14874 + 0. i}
```

Out[30]=



```
In[33]:= A = {{-1, 2, 0}, {2, -4, 0}, {0, 0, 0.1}};
```

```
Eigenvalues[A]
```

```
Eigenvectors[A]
```

```
Out[34]=
```

```
{-5., 0.1, 0.}
```

```
Out[35]=
```

```
{{0.447214, -0.894427, 0.}, {0., 0., 1.}, {0.894427, 0.447214, 0.}}
```